NeoSignal

AI Intelligence Daily · Multi-Source · Cross-Verified

11 May 2026

164 verified · 331 raw · 3 sources

164	1	163	0
TOTAL	VERIFIED	CONFIRMED	EMERGING

Pipeline

Scrape   HackerNews (Show/Top/New), Reddit (r/artificial, r/MachineLearning, r/singularity, r/LocalLLaMA), TechCrunch AI, VentureBeat AI, MIT Tech Review, The Verge AI, Wired AI, ArXiv CS.AI

Filter   40+ AI keyword patterns — LLMs, alignment, safety, research, tooling

Dedup   Title-similarity (45% threshold) merges cross-source variants

Score   Authenticity = 0.5 base + 0.3 per extra source + diversity + HN score bonuses

Rank   Sorted by authenticity. Below 0.25 threshold are dropped.

Authenticity Tiers

VERIFIED	**0.80 - 1.00**	3+ independent sources including media
CONFIRMED	**0.50 - 0.79**	2 sources or 1 high-quality media
EMERGING	**0.25 - 0.49**	Single-source; passes AI keyword filter

Source Breakdown

Source	Articles	Category
ArXiv CS.AI	158	Media
HackerNews	5	Community
TechCrunch AI	2	Media

VERIFIED INTELLIGENCE — 1 story

VERI

Show HN: NanoCorp – Create autonomous companies run by AI

HackerNews +1 source 90% HN 14

arXiv:2510.01685v2 Announce Type: replace-cross Abstract: While large language models (LLMs) appear to be increasingly capable of solving compositional tasks, it is an open question whether they do so using compositional mechanisms. In this work, we investigate how feedforward...

https://www.nanocorp.so

CONFIRMED SIGNALS — 163 stories

CONF

Show HN: adamsreview – better multi-agent PR reviews for Claude Code

HackerNews 50% HN 18

I built adamsreview, a Claude Code plugin that runs deeper, multi-stage PR reviews using parallel sub-agents, validation passes, persistent JSON state, and optional ensemble review via Codex CLI and PR bot comments. On my own PRs, it has been catching dramatically more real bugs...

https://github.com/adamjgmiller/adamsreview

CONF

Show HN: I trained a chess engine to play like humans

HackerNews 50% HN 10

I built 1e4.ai - a chess web app where you play against neural networks trained to mimic human Lichess players at specific Elo ranges. There's a separate model for each 100-point rating bucket from ~800 to 2200+, and the bots not only choose human-like moves but also burn clock...

https://news.ycombinator.com/item?id=48088819

CONF

Show HN: My AI agents bully each other to prevent context drift

HackerNews 50% HN 8

Most multi-agent systems fail the same way: agents drift apart across handoffs. By turn 3 they are working in different realities. By turn 5 they are repeating each other's mistakes and calling it parallelism. WUPHF is an open-source local-first office where AI coworkers run on...

https://wuphf.team

CONF

Show HN: Mosaic – arrange iOS icons by color using an evolutionary algorithm

HackerNews 50% HN 3

It started out as a way for me to freshen up my C++ skills during COVID. But life got in the way and it was put on ice. Luckily, coding LLMs came to the rescue and allowed me to bring it to a point where I feel comfortable sharing it.

https://github.com/RTiK/mosaic

CONF

Anthropic says ‘evil’ portrayals of AI were responsible for Claude’s blackmail attempts

TechCrunch AI 50%

Fictional portrayals of artificial intelligence can have a real effect on AI models, according to Anthropic.

https://techcrunch.com/2026/05/10/anthropic-says-evil-portrayals-of-ai-were-responsible...

CONF

We’re feeling cynical about xAI’s big deal with Anthropic

TechCrunch AI 50%

On the latest episode of the Equity podcast, we discussed what xAI's deal with Anthropic might mean for parent company SpaceX.

https://techcrunch.com/2026/05/10/were-feeling-cynical-about-xais-big-deal-with-anthropic/

CONF

Hidden Coalitions in Multi-Agent AI: A Spectral Diagnostic from Internal Representations

ArXiv CS.AI 50%

arXiv:2605.06696v1 Announce Type: new Abstract: Collections of interacting AI agents can form coalitions, creating emergent group-level organization that is critical for AI safety and alignment. However, observing agent behavior alone is often insufficient to distinguish genuine...

https://arxiv.org/abs/2605.06696

CONF

Saliency-Aware Regularized Quantization Calibration for Large Language Models

ArXiv CS.AI 50%

arXiv:2605.05693v2 Announce Type: replace Abstract: Post-training quantization (PTQ) is an effective approach for deploying large language models (LLMs) under memory and latency constraints. Most existing PTQ methods determine quantization parameters by minimizing a layer-wise...

https://arxiv.org/abs/2605.05693

0 CONF

From Storage to Experience: A Survey on the Evolution of LLM Agent Memory Mechanisms

ArXiv CS.AI 50%

arXiv:2605.06716v1 Announce Type: new Abstract: Large Language Model (LLM)-based agents have fundamentally reshaped artificial intelligence by integrating external tools and planning capabilities. While memory mechanisms have emerged as the architectural cornerstone of these...

https://arxiv.org/abs/2605.06716

1 CONF

When Losses Align: Gradient-Based Composite Loss Weighting for Efficient Pretraining

ArXiv CS.AI 50%

arXiv:2605.07756v1 Announce Type: cross Abstract: Modern deep models are often pretrained on large-scale data with missing labels using composite objectives, where the relative weights of multiple loss terms act as hyperparameters. Tuning these weights with random search or...

https://arxiv.org/abs/2605.07756

2 CONF

Weblica: Scalable and Reproducible Training Environments for Visual Web Agents

ArXiv CS.AI 50%

arXiv:2605.06761v1 Announce Type: new Abstract: The web is complex, open-ended, and constantly changing, making it challenging to scale training data for visual web agents. Existing data collection attempts remain limited to offline trajectories for supervised fine-tuning or a...

https://arxiv.org/abs/2605.06761

3 CONF

When Does Critique Improve AI-Assisted Theoretical Physics? SCALAR: Structured Critic--Actor Loop for Agentic Reasoning

ArXiv CS.AI 50%

arXiv:2605.06772v1 Announce Type: new Abstract: As large language models (LLMs) show increasing promise on research-level physics reasoning tasks and agentic AI becomes more common, a practical question emerges: How does the interaction between researchers and agents affect the...

https://arxiv.org/abs/2605.06772

4 CONF

Towards Security-Auditable LLM Agents: A Unified Graph Representation

ArXiv CS.AI 50%

arXiv:2605.06812v1 Announce Type: new Abstract: LLM-based agentic systems are rapidly evolving to perform complex autonomous tasks through dynamic tool invocation, stateful memory management, and multi-agent collaboration. However, this semantics-driven execution paradigm...

https://arxiv.org/abs/2605.06812

5 CONF

TimeLesSeg: Unified Contrast-Agnostic Cross-Sectional and Longitudinal MS Lesion Segmentation via a Stochastic Generative Model

ArXiv CS.AI 50%

arXiv:2605.07955v1 Announce Type: cross Abstract: Multiple sclerosis (MS) expresses substantial clinical and radiological heterogeneity, which poses significant challenges for automatic lesion segmentation. The current deep learning-based SOTA is highly susceptible to changes in...

https://arxiv.org/abs/2605.07955

6 CONF

Replicating Human Motivated Reasoning Studies with LLMs

ArXiv CS.AI 50%

arXiv:2601.16130v2 Announce Type: replace-cross Abstract: Motivated reasoning - the idea that individuals processing information may be motivated to either arrive at accurate beliefs or arrive at desired conclusions - has been well-explored as a human phenomenon. However, it...

https://arxiv.org/abs/2601.16130

7 CONF

MathlibPR: Pull Request Merge-Readiness Benchmark for Formal Mathematical Libraries

ArXiv CS.AI 50%

arXiv:2605.07147v1 Announce Type: cross Abstract: The ecosystem of Lean and Mathlib has become the de facto standard for large language model (LLM) assisted formal reasoning with remarkable successes in recent years. Those successes, however, only consume Mathlib as an essential...

https://arxiv.org/abs/2605.07147

8 CONF

How Well Do LLMs Perform on the Simplest Long-Chain Reasoning Tasks: An Empirical Study on the Equivalence Class Problem

ArXiv CS.AI 50%

arXiv:2605.06882v1 Announce Type: new Abstract: Large Language Models (LLMs) have achieved great improvements in recent years. Nevertheless, it still remains unclear how good LLMs are for reasoning tasks, especially for long-chain ones. In this paper, we evaluate LLMs'...

https://arxiv.org/abs/2605.06882

9 CONF

Beyond the Black Box: Interpretability of Agentic AI Tool Use

ArXiv CS.AI 50%

arXiv:2605.06890v1 Announce Type: new Abstract: AI agents are promising for high-stakes enterprise workflows, but dependable deployment remains limited because tool-use failures are difficult to diagnose and control. Agents may skip required tool calls, invoke tools...

https://arxiv.org/abs/2605.06890

0 CONF

ESSAM: A Novel Competitive Evolution Strategies Approach to Reinforcement Learning for Memory Efficient LLMs Fine-Tuning

ArXiv CS.AI 50%

arXiv:2602.01003v2 Announce Type: replace-cross Abstract: Reinforcement learning (RL) has become a key training step for improving mathematical reasoning in large language models (LLMs), but it often has high GPU memory usage, which makes it hard to use in settings with limited...

https://arxiv.org/abs/2602.01003

1 CONF

Hierarchical Task Network Planning with LLM-Generated Heuristics

ArXiv CS.AI 50%

arXiv:2605.07707v1 Announce Type: new Abstract: HTN planning is a variation of classical planning where, instead of searching for a linear sequence of actions, an algorithm decomposes higher-level tasks using a method library until only executable actions remain. On one hand,...

https://arxiv.org/abs/2605.07707

2 CONF

Adaptive auditing of AI systems with anytime-valid guarantees

ArXiv CS.AI 50%

arXiv:2605.07002v1 Announce Type: new Abstract: A major bottleneck in characterizing the failure modes of generative AI systems is the cost and time of annotation and evaluation. Consequently, adaptive testing paradigms have gained popularity, where one opportunistically decides...

https://arxiv.org/abs/2605.07002

3 CONF

2.5-D Decomposition for LLM-Based Spatial Construction

ArXiv CS.AI 50%

arXiv:2605.07066v1 Announce Type: new Abstract: Autonomous systems that build structures from natural-language instructions need reliable spatial reasoning, yet large language models (LLMs) make systematic coordinate errors when generating three-dimensional block placements. We...

https://arxiv.org/abs/2605.07066

4 CONF

Switchcraft: AI Model Router for Agentic Tool Calling

ArXiv CS.AI 50%

arXiv:2605.07112v1 Announce Type: new Abstract: Agentic AI systems that invoke external tools are powerful but costly, leading developers to default to large models and overspend inference budgets. Model routing can mitigate this, but existing routers are designed for chat...

https://arxiv.org/abs/2605.07112

5 CONF

Can You Break RLVER? Probing Adversarial Robustness of RL-Trained Empathetic Agents

ArXiv CS.AI 50%

arXiv:2605.07138v1 Announce Type: new Abstract: Reinforcement learning from verifiable emotion rewards RLVER has produced language models with strong empathetic performance, evaluated on benchmarks that assume cooperative, honest users. Yet real emotional interactions...

https://arxiv.org/abs/2605.07138

6 **CONF**

SREGym: A Live Benchmark for AI SRE Agents with High-Fidelity Failure Scenarios

ArXiv CS.AI **50%**

arXiv:2605.07161v1 Announce Type: new Abstract: AI agents are increasingly used to diagnose and mitigate failures in production systems, known as agentic Site Reliability Engineering (SRE). Current SRE benchmarks are limited to oversimplistic SRE tasks and are unfortunately hard...

https://arxiv.org/abs/2605.07161

7 **CONF**

Repeated Deceptive Path Planning against Learnable Observer

ArXiv CS.AI **50%**

arXiv:2605.07174v1 Announce Type: new Abstract: We study the problem of deceptive path planning (DPP), where an agent aims to conceal its true destination from external observers. While existing work assumes static, non-learning observers, real-world adversaries-such as in...

https://arxiv.org/abs/2605.07174

8 **CONF**

Three-in-One World Model: Energy-Based Consistency, Prediction, and Counterfactual Inference for Marketing Intervention

ArXiv CS.AI **50%**

arXiv:2605.07199v1 Announce Type: new Abstract: Marketing decisions reflect the interaction of latent consumer heterogeneity, time-varying internal states, and explicit interventions, a structure that current prediction- and language-oriented models do not capture in a unified...

https://arxiv.org/abs/2605.07199

9 **CONF**

MEMOREPAIR: Barrier-First Cascade Repair in Agentic Memory

ArXiv CS.AI **50%**

arXiv:2605.07242v1 Announce Type: new Abstract: Agentic memory evolves across tasks into durable derived artifacts: summaries, cached outputs, embeddings, learned skills, and executable tool procedures. When a source artifact is deleted, corrected, or invalidated by tool or API...

https://arxiv.org/abs/2605.07242

0 **CONF**

Can Agents Price a Reaction? Evaluating LLMs on Chemical Cost Reasoning

ArXiv CS.AI **50%**

arXiv:2605.07251v1 Announce Type: new Abstract: Large Language Models (LLMs) have become increasingly capable as tool-using agents, with benchmarks spanning diverse general agentic tasks. Yet rigorous evaluation of scientific tool use remains limited. In chemistry, recent agents...

https://arxiv.org/abs/2605.07251

1 **CONF**

Structured Role-Aware Policy Optimization for Multimodal Reasoning

ArXiv CS.AI **50%**

arXiv:2605.07274v1 Announce Type: new Abstract: Reinforcement learning from verifiable rewards (RLVR), especially with Group Relative Policy Optimization (GRPO), has shown strong potential for improving the reasoning capabilities of large vision-language models (LVLMs). However,...

https://arxiv.org/abs/2605.07274

2 **CONF**

Signal Reshaping for GRPO in Weak-Feedback Agentic Code Repair

ArXiv CS.AI **50%**

arXiv:2605.07276v1 Announce Type: new Abstract: Code-agent RL often receives weak feedback: rollout-time signals are reliable and executable, but capture only necessary or surface conditions for task success rather than the target semantic predicate. Using agentic compile-fix as...

https://arxiv.org/abs/2605.07276

3 **CONF**

Dr. Post-Training: A Data Regularization Perspective on LLM Post-Training

ArXiv CS.AI **50%**

arXiv:2605.07063v1 Announce Type: cross Abstract: Data selection methods address a critical challenge in LLM post-training: effectively leveraging scarce, high-fidelity target data alongside abundant but imperfectly aligned general training data. In this work, we move beyond the...

https://arxiv.org/abs/2605.07063

4 **CONF**

Discovering Ordinary Differential Equations with LLM-Based Qualitative and Quantitative Evaluation

ArXiv CS.AI **50%**

arXiv:2605.07323v1 Announce Type: new Abstract: Discovering governing differential equations from observational data is a fundamental challenge in scientific machine learning. Existing symbolic regression approaches rely primarily on quantitative metrics; however, real-world...

https://arxiv.org/abs/2605.07323

5 **CONF**

Inference Time Causal Probing in LLMs

ArXiv CS.AI **50%**

arXiv:2605.07631v1 Announce Type: new Abstract: Causal probing methods aim to test and control how internal representations influence the behavior of generative models. In causal probing, an intervention modifies hidden states so that a property takes on a different value. Most...

https://arxiv.org/abs/2605.07631

6 **CONF**

MISA: Mixture of Indexer Sparse Attention for Long-Context LLM Inference

ArXiv CS.AI **50%**

arXiv:2605.07363v1 Announce Type: cross Abstract: DeepSeek Sparse Attention (DSA) sets the state of the art for fine-grained inference-time sparse attention by introducing a learned token-wise indexer that scores every prefix token and selects the most relevant ones for the main...

https://arxiv.org/abs/2605.07363

7 **CONF**

Efficient Data Selection for Multimodal Models via Incremental Optimization Utility

ArXiv CS.AI **50%**

arXiv:2605.07488v1 Announce Type: new Abstract: The scaling of Large Multimodal Models (LMMs) is constrained by the quality-quantity trade-off inherent in synthetic data. Previous approaches, such as LLM-as-a-Judge, have proven their effectiveness in addressing this but suffer...

https://arxiv.org/abs/2605.07488

8 CONF

ProteinJEPA: Latent prediction complements protein language models

ArXiv CS.AI 50%

arXiv:2605.07554v1 Announce Type: cross Abstract: Protein language models are trained primarily with masked language modeling (MLM), which predicts amino-acid identities at masked positions. We ask whether latent-space prediction can complement these token-level objectives under...

https://arxiv.org/abs/2605.07554

9 CONF

Parallel Lifted Planning via Semi-Naive Datalog Evaluation

ArXiv CS.AI 50%

arXiv:2605.07584v1 Announce Type: new Abstract: Lifted classical planners operate directly on first-order planning tasks to avoid the computationally demanding grounding step. However, lifted planning is typically slower, as planners must repeatedly instantiate ground structures...

https://arxiv.org/abs/2605.07584

0 CONF

TCDA: Thread-Constrained Discourse-Aware Modeling for Conversational Sentiment Quadruple Analysis

ArXiv CS.AI 50%

arXiv:2605.01717v2 Announce Type: replace-cross Abstract: Conversational Aspect-based Sentiment Quadruple Analysis (DiaASQ) needs to capture the complex interrelationships in multiple rounds of dialogues. Existing methods usually employ simple Graph Convolutional Networks (GCN),...

https://arxiv.org/abs/2605.01717

1 CONF

TraceFix: Repairing Agent Coordination Protocols with TLA+ Counterexamples

ArXiv CS.AI 50%

arXiv:2605.07935v1 Announce Type: new Abstract: We present TraceFix, a verification-first pipeline for Large Language Model (LLM) multi-agent coordination. An agent synthesizes a protocol topology as a structured intermediate representation (IR) from a task description,...

https://arxiv.org/abs/2605.07935

2 CONF

The Limits of AI-Driven Allocation: Optimal Screening under Aleatoric Uncertainty

ArXiv CS.AI 50%

arXiv:2605.07979v1 Announce Type: new Abstract: The rise of machine learning has shifted targeted resource allocation in policy and humanitarian settings toward algorithmic targeting based on predicted risk scores. This approach is typically cheaper and faster than traditional...

https://arxiv.org/abs/2605.07979

3 CONF

Reason to Play: Behavioral and Brain Alignment Between Frontier LRMs and Human Game Learners

ArXiv CS.AI 50%

arXiv:2605.08019v1 Announce Type: new Abstract: Humans rapidly learn abstract knowledge when encountering novel environments and flexibly deploy this knowledge to guide efficient and intelligent action. Can modern AI systems learn and plan in a similar way? We study this...

https://arxiv.org/abs/2605.08019

4 CONF

CommFuse: Hiding Tail Latency via Communication Decomposition and Fusion for Distributed LLM Training

ArXiv CS.AI 50%

arXiv:2604.24013v2 Announce Type: cross Abstract: The rapid growth in the size of large language models has necessitated the partitioning of computational workloads across accelerators such as GPUs, TPUs, and NPUs. However, these parallelization strategies incur substantial data...

https://arxiv.org/abs/2604.24013

5 CONF

Evaluating Prompt Injection Defenses for Educational LLM Tutors: Security-Usability-Latency Trade-offs

ArXiv CS.AI 50%

arXiv:2605.06669v1 Announce Type: cross Abstract: Educational LLM tutors face a core AI alignment challenge: they must follow user intent while preserving pedagogical constraints and safety policies. We present an evaluation methodology for prompt-injection defenses in this...

https://arxiv.org/abs/2605.06669

6 CONF

Domain-level metacognitive monitoring in frontier LLMs: A 33-model atlas

ArXiv CS.AI 50%

arXiv:2605.06673v1 Announce Type: cross Abstract: Aggregate metacognitive quality scores mask within-model variation across MMLU benchmark domains. We administered 1,500 MMLU items (250 per domain, under an a priori six-domain grouping) to 33 frontier LLMs from eight model...

https://arxiv.org/abs/2605.06673

7 CONF

Multimodal synthesis of MRI and tabular data with diffusion in a joint latent space via cross-attention

ArXiv CS.AI 50%

arXiv:2605.06699v1 Announce Type: cross Abstract: We propose a multimodal latent diffusion model that jointly synthesizes volumetric magnetic resonance imaging (MRI) and tabular clinical data within a shared latent space via cross-attention. This approach enables coherent joint...

https://arxiv.org/abs/2605.06699

8 CONF

The Single-File Test: A Longitudinal Public-Interface Evaluation of First-Output LLM Web Generation with Social Reach Tracking

ArXiv CS.AI 50%

arXiv:2605.06707v1 Announce Type: cross Abstract: This paper presents an eight-week observational comparison of 68 single-file HTML generations collected across 17 public experiments in the "HTML AI Battle" project between December 10, 2025 and February 4, 2026. Four reasoning...

https://arxiv.org/abs/2605.06707

9 CONF

Agentic AI and the Industrialization of Cyber Offense: Forecast, Consequences, and Defensive Priorities for Enterprises and the Mittelstand

ArXiv CS.AI 50%

arXiv:2605.06713v1 Announce Type: cross Abstract: Agentic AI systems can plan, call tools, inspect code, interact with web applications, and coordinate multi-step workflows. These same capabilities change the economics of cyber offense. The central near-term risk is not that...

https://arxiv.org/abs/2605.06713

0 CONF

Conditional generation of antibody sequences with classifier-guided germline-absorbing discrete diffusion

ArXiv CS.AI 50%

arXiv:2605.06720v1 Announce Type: cross Abstract: Antibody therapeutics are among the most successful modern medicines, yet computationally designing antibodies with desirable binding and developability properties remains challenging. While protein language models (pLMs) have...

https://arxiv.org/abs/2605.06720

1 CONF

Enabling Unsupervised Training of Deep EEG Denoisers With Intelligent Partitioning

ArXiv CS.AI 50%

arXiv:2605.06724v1 Announce Type: cross Abstract: Denoising wearable electroencephalogram (EEG) is inherently challenging since neural activity is not only subtle but also inseparable from spectrally overlapping noise artifacts. Classical signal processing methods, relying on...

https://arxiv.org/abs/2605.06724

2 CONF

The EΔ-MHC-Geo Transformer: Adaptive Geodesic Operations with Guaranteed Orthogonality

ArXiv CS.AI 50%

arXiv:2605.06729v1 Announce Type: cross Abstract: We present the EΔ-MHC-Geo Transformer, a novel architecture that unifies Manifold-Constrained Hyper-Connections (mHC), Deep Delta Learning (DDL), and the Cayley transform to obtain input-adaptive, unconditionally...

https://arxiv.org/abs/2605.06729

3 CONF

An Interpretable and Scalable Framework for Evaluating Large Language Models

ArXiv CS.AI 50%

arXiv:2605.07046v1 Announce Type: cross Abstract: Evaluation of large language models (LLMs) is increasingly critical, yet standard benchmarking methods rely on average accuracy, overlooking both the inherent stochasticity of LLM outputs and the heterogeneity of benchmark items....

https://arxiv.org/abs/2605.07046

4 CONF

From Specification to Deployment: Empirical Evidence from a W3C VC + DID Trust Infrastructure for Autonomous Agents

ArXiv CS.AI 50%

arXiv:2605.06738v1 Announce Type: cross Abstract: Autonomous AI agents now transact at production scale -- 69,000 bots executing 165 million transactions across 50 million USDC in cumulative volume on a single marketplace -- without any shared trust layer between participants....

https://arxiv.org/abs/2605.06738

5 CONF

A Linear-Transformer Hybrid for SNP-Based Genotype-to-Phenotype Prediction in Grapevine

ArXiv CS.AI 50%

arXiv:2605.06762v1 Announce Type: cross Abstract: Robust genotype-to-phenotype (G2P) prediction is essential for accelerating breeding decisions and genetic gain. However, it remains challenging to measure complex traits under variable field conditions and across years. In this...

https://arxiv.org/abs/2605.06762

6 **CONF**

PAMPOS: Causal Transformer-based Trajectory Prediction for Attack-Agnostic Misbehavior Detection in V2X Networks

ArXiv CS.AI **50%**

arXiv:2605.06833v1 Announce Type: cross Abstract: Misbehavior detection in Vehicle-to-Everything (V2X) networks is a second line of defense against insider falsification attacks that cryptographic mechanisms alone cannot address. Existing learning-based Misbehavior Detection...

https://arxiv.org/abs/2605.06833

7 **CONF**

On Privacy Leakage in Tabular Diffusion Models: Influential Factors, Attacker Knowledge, and Metrics

ArXiv CS.AI **50%**

arXiv:2605.06835v1 Announce Type: cross Abstract: Tabular data plays an important role in many fields and industries, including those with elevated privacy considerations and risks. As such, there is a rising interest in generating high-quality synthetic proxies for real tabular...

https://arxiv.org/abs/2605.06835

8 **CONF**

LLM-Guided Open Hypothesis Learning from Autonomous Scanning Probe Microscopy Experiments

ArXiv CS.AI **50%**

arXiv:2605.06839v1 Announce Type: cross Abstract: Autonomous experimentation has transformed microscopy and materials discovery by enabling closed-loop optimization including imaging and spectroscopy tuning, strucutre property relationship discovery, and exploration of...

https://arxiv.org/abs/2605.06839

9 **CONF**

How to Compress KV Cache in RL Post-Training? Shadow Mask Distillation for Memory-Efficient Alignment

ArXiv CS.AI **50%**

arXiv:2605.06850v1 Announce Type: cross Abstract: Reinforcement Learning (RL) has emerged as a crucial paradigm for unlocking the advanced reasoning capabilities of Large Language Models (LLMs), encompassing frameworks like RLHF and RLAIF. Regardless of the specific optimization...

https://arxiv.org/abs/2605.06850

0 **CONF**

Don't Retrain, Align: Adapting Autoregressive LMs to Diffusion LMs via Representation Alignment

ArXiv CS.AI **50%**

arXiv:2605.06885v1 Announce Type: cross Abstract: Diffusion language models (DLMs) have recently demonstrated capabilities that complement standard autoregressive (AR) models, particularly in non-sequential generation and bidirectional editing. Although recent work has shown...

https://arxiv.org/abs/2605.06885

1 **CONF**

MIST: Multimodal Interactive Speech-based Tool-calling Conversational Assistants for Smart Homes

ArXiv CS.AI **50%**

arXiv:2605.06897v1 Announce Type: cross Abstract: The rise of Internet of Things (IoT) devices in the physical world necessitates voice-based interfaces capable of handling complex user experiences. While modern Large Language Models (LLMs) already demonstrate strong tool-usage...

https://arxiv.org/abs/2605.06897

2 **CONF**

MELD: Multi-Task Equilibrated Learning Detector for AI-Generated Text

ArXiv CS.AI **50%**

arXiv:2605.06903v1 Announce Type: cross Abstract: Large language models are now embedded in everyday writing workflows, making reliable AI-generated text detection important for academic integrity, content moderation, and provenance tracking. In practice, however, a detector...

https://arxiv.org/abs/2605.06903

3 **CONF**

Same Signal, Opposite Meaning: Direction-Informed Adaptive Learning for LLM Agents

ArXiv CS.AI **50%**

arXiv:2605.06908v1 Announce Type: cross Abstract: Adaptive test-time compute for LLM agents aims to invoke extra computation only when it improves performance. Existing methods typically use confidence-, uncertainty-, or difficulty-based gates, assuming a fixed direction from...

https://arxiv.org/abs/2605.06908

4 **CONF**

Reformulating KV Cache Eviction Problem for Long-Context LLM Inference

ArXiv CS.AI **50%**

arXiv:2605.07234v1 Announce Type: cross Abstract: Large language models (LLMs) support long-context inference but suffer from substantial memory and runtime overhead due to Key-Value (KV) Cache growth. Existing KV Cache eviction methods primarily rely on local attention weights,...

https://arxiv.org/abs/2605.07234

5 **CONF**

Bridging the Last Mile of Circuit Design: PostEDA-Bench, a Hierarchical Benchmark for PPA Convergence and DRC Fixing

ArXiv CS.AI **50%**

arXiv:2605.06936v1 Announce Type: cross Abstract: LLM-based agents are increasingly applied to the "last mile" of Electronic Design Automation (EDA): repairing residual sign-off Design Rule Check (DRC) violations and converging Power-Performance-Area (PPA) targets after tool...

https://arxiv.org/abs/2605.06936

6 **CONF**

A Generalized Singular Value Theory for Neural Networks

ArXiv CS.AI **50%**

arXiv:2605.06938v1 Announce Type: cross Abstract: Building on the abstract Generalized Singular Value Decomposition (GSVD) theory of Brown et al. [2025], we prove that most modern neural architectures admit a generalized SVD representation in which they are left-invertible...

https://arxiv.org/abs/2605.06938

7 **CONF**

From Surface Learning to Deep Understanding: A Grounded AI Tutoring System for Moodle

ArXiv CS.AI **50%**

arXiv:2605.06963v1 Announce Type: cross Abstract: This demo paper describes the development of the AI Teaching \& Learning Assistant, a modular Moodle plugin that leverages Retrieval-Augmented Generation (RAG) to deliver high-quality, hallucination-free education. The system...

https://arxiv.org/abs/2605.06963

8 CONF

AI and Consciousness: Shifting Focus Towards Tractable Questions

ArXiv CS.AI 50%

arXiv:2605.06965v1 Announce Type: cross Abstract: As language-based AI systems become more anthropomorphic, the question of whether they can have subjective experience is increasingly pressing. I focus here on the tractability of research questions in the space of AI...

https://arxiv.org/abs/2605.06965

9 CONF

TAP: Two-Stage Adaptive Personalization of Multi-Task and Multi-Modal Foundation Models in Federated Learning

ArXiv CS.AI 50%

arXiv:2509.26524v3 Announce Type: replace-cross Abstract: In federated learning (FL), local personalization of models has received significant attention, yet personalized fine-tuning of foundation models remains underexplored. In particular, there is a lack of understanding in...

https://arxiv.org/abs/2509.26524

0 CONF

BGM-IV: an AI-powered Bayesian generative modeling approach for instrumental variable analysis

ArXiv CS.AI 50%

arXiv:2605.07029v1 Announce Type: cross Abstract: Instrumental-variable (IV) regression enables causal estimation under endogeneity, but modern IV problems often involve nonlinear structural effects and high-dimensional covariates. Existing nonlinear IV methods directly learn...

https://arxiv.org/abs/2605.07029

1 CONF

A Geometric Taxonomy of Hallucinations in LLMs

ArXiv CS.AI 50%

arXiv:2602.13224v3 Announce Type: replace Abstract: Hallucinations in deployed language models can have real consequences for downstream decisions in domains such as healthcare, legal, and financial services. In production, detection has to run on what the deployed system can...

https://arxiv.org/abs/2602.13224

2 CONF

Unlocking High-Fidelity Molecular Generation from Mass Spectra via Dual-Stream Line Graph Diffusion

ArXiv CS.AI 50%

arXiv:2605.07048v1 Announce Type: cross Abstract: De novo molecular generation from tandem mass spectra is a challenging inverse problem whose core difficulty lies in the circular dependency between atom-level and bond-level reasoning: determining a bond's type requires knowing...

https://arxiv.org/abs/2605.07048

3 CONF

Pan-FM: A Pan-Organ Foundation Model with Saliency-Guided Masking for Missing Robustness

ArXiv CS.AI 50%

arXiv:2605.07055v1 Announce Type: cross Abstract: Foundation models (FMs) have shown great promise in medical imaging, but most FMs are trained on unimodal data within isolated domains, such as brain MRI alone. Human aging and disease arise through coordinated biological...

https://arxiv.org/abs/2605.07055

4 CONF

MemSearcher: Training LLMs to Reason, Search and Manage Memory via End-to-End Reinforcement Learning

ArXiv CS.AI 50%

arXiv:2511.02805v2 Announce Type: replace-cross Abstract: LLM-based search agents often concatenate the full interaction history into the context, producing long and noisy inputs, and increasing compute cost and GPU memory overhead. To address this issue, we propose MemSearcher,...

https://arxiv.org/abs/2511.02805

5 CONF

Causal EpiNets: Precision-corrected Bounds on Individual Treatment Effects using Epistemic Neural Networks

ArXiv CS.AI 50%

arXiv:2605.07065v1 Announce Type: cross Abstract: Individual treatment effects are not point-identified from data. The Probability of Necessity and Sufficiency (PNS) circumvents this limitation by characterizing individual-level causality through intersection bounds derived from...

https://arxiv.org/abs/2605.07065

6 CONF

WiCER: Wiki-memory Compile, Evaluate, Refine Iterative Knowledge Compilation for LLM Wiki Systems

ArXiv CS.AI 50%

arXiv:2605.07068v1 Announce Type: cross Abstract: The LLM Wiki pattern, to compile and provide domain knowledge into a persistent artifact and serve it to LLMs via KV cache inference, promises context access at sub-second latency with zero retrieval failure. Realizing this...

https://arxiv.org/abs/2605.07068

7 CONF

The Translation Tax Is Not a Scalar: A Counterfactual Audit of English-Source Cue Inheritance in Chinese Multilingual Benchmarks

ArXiv CS.AI 50%

arXiv:2605.07093v1 Announce Type: cross Abstract: The Translation Tax is often treated as a scalar: translated benchmarks are assumed to inflate scores by preserving English-source cues. We audit this claim in an English-to-Chinese setting. Three proxy estimators disagree:...

https://arxiv.org/abs/2605.07093

8 CONF

Stabilized neural Hamilton--Jacobi--Bellman solvers: Error analysis and applications in model-based reinforcement learning

ArXiv CS.AI 50%

arXiv:2605.07116v1 Announce Type: cross Abstract: Physics-informed neural solvers offer a promising route to model-based reinforcement learning in continuous time, where optimal feedback synthesis is governed by Hamilton--Jacobi--Bellman (HJB) equations. Practical...

https://arxiv.org/abs/2605.07116

9 CONF

An Embarrassingly Simple Graph Heuristic Reveals Shortcut-Solvable Benchmarks for Sequential Recommendation

ArXiv CS.AI 50%

arXiv:2605.07125v1 Announce Type: cross Abstract: Sequential recommendation has increasingly shifted toward generative recommenders that combine sequential patterns with semantic item information. Yet these methods are often evaluated on a small set of widely used benchmarks,...

https://arxiv.org/abs/2605.07125

0 CONF

DCGL: Dual-Channel Graph Learning with Large Language Models for Knowledge-Aware Recommendation

ArXiv CS.AI 50%

arXiv:2605.07314v1 Announce Type: cross Abstract: Knowledge Graphs (KGs) have proven highly effective for recommendation systems by capturing latent item relationships, while recent integration of Large Language Models (LLMs) has further enhanced semantic understanding and...

https://arxiv.org/abs/2605.07314

1 CONF

GAD in the Wild: Benchmarking Graph Anomaly Detection under Realistic Deployment Challenges

ArXiv CS.AI 50%

arXiv:2605.07133v1 Announce Type: cross Abstract: Graph Anomaly Detection (GAD) is a critical task in graph machine learning with vital applications in financial fraud detection and social platform governance. However, existing GAD benchmarks are often restricted to small-scale,...

https://arxiv.org/abs/2605.07133

2 CONF

Adaptive Negative Reinforcement for LLM Reasoning:Dynamically Balancing Correction and Diversity in RLVR

ArXiv CS.AI 50%

arXiv:2605.07137v1 Announce Type: cross Abstract: Reinforcement learning with verifiable rewards (RLVR) has become a highly effective method for improving the reasoning abilities of Large Language Models (LLMs). Recent research shows that Negative Sample Reinforcement (NSR) --...

https://arxiv.org/abs/2605.07137

3 CONF

RELO: Reinforcement Learning to Localize for Visual Object Tracking

ArXiv CS.AI 50%

arXiv:2605.07379v1 Announce Type: cross Abstract: Conventional visual object trackers localize targets using handcrafted spatial priors, often in the form of heatmaps. Such priors provide only surrogate supervision and are poorly aligned with tracking optimization and evaluation...

https://arxiv.org/abs/2605.07379

4 CONF

Persistent Visual Memory: Sustaining Perception for Deep Generation in LVLMs

ArXiv CS.AI 50%

arXiv:2605.00814v2 Announce Type: replace-cross Abstract: While autoregressive Large Vision-Language Models (LVLMs) demonstrate remarkable proficiency in multimodal tasks, they face a "Visual Signal Dilution" phenomenon, where the accumulation of textual history expands the...

https://arxiv.org/abs/2605.00814

5 CONF

Closed-Form Linear-Probe Dataset Distillation for Pre-trained Vision Models

ArXiv CS.AI 50%

arXiv:2605.07194v1 Announce Type: cross Abstract: Dataset distillation compresses a large training set into a small synthetic set that preserves downstream training utility. While most existing methods target training networks from scratch, modern visual transfer learning often...

https://arxiv.org/abs/2605.07194

6 CONF

PSK@EEUCA 2026: Fine-Tuning Large Language Models with Synthetic Data Augmentation for Multi-Class Toxicity Detection in Gaming Chat

ArXiv CS.AI 50%

arXiv:2605.07201v1 Announce Type: cross Abstract: This paper describes our system for the EEUCA 2026 Shared Task on Understanding Toxic Behavior in Gaming Communities. The task involves classifying World of Tanks chat messages into six toxicity categories: Non-toxic,...

https://arxiv.org/abs/2605.07201

7 CONF

Same Brain, Different Prediction: How Preprocessing Choices Undermine EEG Decoding Reliability

ArXiv CS.AI 50%

arXiv:2605.07212v1 Announce Type: cross Abstract: Electroencephalography (EEG) is a cornerstone of brain-computer interfaces and clinical neuroscience, yet deep learning models are typically trained and evaluated under a single, unreported preprocessing pipeline. We formalize...

https://arxiv.org/abs/2605.07212

8 CONF

Hard to Read, Easy to Jailbreak: How Visual Degradation Bypasses MLLM Safety Alignment

ArXiv CS.AI 50%

arXiv:2605.07250v1 Announce Type: cross Abstract: Recent advancements in visual context compression enable MLLMs to process ultra-long contexts efficiently by rendering text into images. However, we identify a critical vulnerability inherent to this paradigm: lowering image...

https://arxiv.org/abs/2605.07250

9 CONF

Resource-Element Energy Difference for Noncoherent Over-the-Air Federated Learning

ArXiv CS.AI 50%

arXiv:2605.07263v1 Announce Type: cross Abstract: Over-the-air federated learning (OTA-FL) reduces uplink latency by exploiting waveform superposition, but conventional analog aggregation schemes typically require instantaneous channel state information (CSI), channel inversion,...

https://arxiv.org/abs/2605.07263

0 CONF

Vaporizer: Breaking Watermarking Schemes for Large Language Model Outputs

ArXiv CS.AI 50%

arXiv:2605.07481v1 Announce Type: cross Abstract: In this paper, we investigate the recent state-of-the-art schemes for watermarking large language models (LLMs) outputs. These techniques are claimed to be robust, scalable and production-grade, aimed at promoting responsible...

https://arxiv.org/abs/2605.07481

1 CONF

From Clouds to Hallucinations: Atmospheric Retrieval Hijacking in Remote Sensing Vision-Language RAG

ArXiv CS.AI 50%

arXiv:2605.07273v1 Announce Type: cross Abstract: Multimodal RAG systems increasingly rely on vision-language retrievers to ground visual queries in external textual evidence. Existing adversarial studies on RAG mainly manipulate the retrieval corpus or memory, while attacks on...

https://arxiv.org/abs/2605.07273

2 CONF

Mask2Cause: Causal Discovery via Adjacency Constrained Causal Attention

ArXiv CS.AI 50%

arXiv:2605.07280v1 Announce Type: cross Abstract: Leveraging deep learning for causal discovery in time series remains challenging because existing neural methods predominantly rely on component-wise architectures that fail to capture shared system dynamics or employ decoupled...

https://arxiv.org/abs/2605.07280

3 CONF

LoopNav: Benchmarking Spatial Consistency in World Models

ArXiv CS.AI 50%

arXiv:2505.22976v3 Announce Type: replace-cross Abstract: The ability to simulate the world in a spatially consistent manner is a crucial requirement for effective world models. Such a model enables high-quality visual generation, and also ensures the reliability of world models...

https://arxiv.org/abs/2505.22976

4 CONF

Generative Modeling with Flux Matching

ArXiv CS.AI 50%

arXiv:2605.07319v1 Announce Type: cross Abstract: We introduce Flux Matching, a new paradigm for generative modeling that generalizes existing score-based models to a broader family of vector fields that need not be conservative. Rather than requiring the model to equal the data...

https://arxiv.org/abs/2605.07319

5 CONF

Mage: Multi-Axis Evaluation of LLM-Generated Executable Game Scenes Beyond Compile-Pass Rate

ArXiv CS.AI 50%

arXiv:2605.07342v1 Announce Type: cross Abstract: Compile-pass rate is the dominant evaluation signal for LLM code generation, yet for multi-component domain-specific artifacts it can be actively misleading. We demonstrate this on executable game scene synthesis with a four-axis...

https://arxiv.org/abs/2605.07342

6 CONF

OrchJail: Jailbreaking Tool-Calling Text-to-Image Agents by Orchestration-Guided Fuzzing

ArXiv CS.AI 50%

arXiv:2605.07414v1 Announce Type: cross Abstract: Tool-calling text-to-image (T2I) agents can plan and execute multi-step tool chains to accomplish complex generation and editing queries. However, this capability introduces a new safety attack surface: harmful outputs may arise...

https://arxiv.org/abs/2605.07414

7 CONF

Prompt Engineering Strategies for LLM-based Qualitative Coding of Psychological Safety in Software Engineering Communities: A Controlled Empirical Study

ArXiv CS.AI 50%

arXiv:2605.07422v1 Announce Type: cross Abstract: Qualitative analysis plays a pivotal role in understanding the human and social aspects of software engineering. However, it remains a demanding process shaped by the subjective interpretation of individual researchers and...

https://arxiv.org/abs/2605.07422

8 CONF

Physical Simulators as Do-Operators: Causal Discovery under Latent Confounders for AI-for-Science

ArXiv CS.AI 50%

arXiv:2605.07467v1 Announce Type: cross Abstract: Existing interventional causal discovery methods -- IGSP, DCDI, ENCO -- assume causal sufficiency (no latent confounders) and rely on virtual interventions in synthetic simulators. In AI-for-Science settings such as molecular...

https://arxiv.org/abs/2605.07467

9 CONF

HBEE: Human Behavioral Entropy Engine -- Pre-Registered Multi-Agent LLM Simulation of Peer-Suspicion-Based Detection Inversion

ArXiv CS.AI 50%

arXiv:2605.07472v1 Announce Type: cross Abstract: Insider threat detection assumes that an adaptive insider leaves behavioral residue distinguishing them from legitimate users. We test this assumption against an LLM-driven adaptive insider in a controlled multi-agent simulator....

https://arxiv.org/abs/2605.07472

00 CONF

SHRED: Retain-Set-Free Unlearning via Self-Distillation with Logit Demotion

ArXiv CS.AI 50%

arXiv:2605.07482v1 Announce Type: cross Abstract: Machine unlearning for large language models (LLMs) aims to selectively remove memorized content such as private data, copyrighted text, or hazardous knowledge, without costly full retraining. Most existing methods require a...

https://arxiv.org/abs/2605.07482

01 CONF

Does Your Neural Network Extrapolate? Feature Engineering as Identifiability Bias for OOD Generalization

ArXiv CS.AI 50%

arXiv:2605.07483v1 Announce Type: cross Abstract: Successful deep neural networks discover salient features of data. We show when and why they fail to learn out-of-distribution (OOD)-relevant representations from an in-distribution (ID) training window. This requires decoupling...

https://arxiv.org/abs/2605.07483

02 CONF

Excluding the Target Domain Improves Extrapolation: Deconfounded Hierarchical Physics Constraints

ArXiv CS.AI 50%

arXiv:2605.07485v1 Announce Type: cross Abstract: Extrapolation to out-of-distribution conditions is a fundamental challenge for physics-constrained deep generative models. Existing methods apply physical constraints as a single static regularization term uniformly across the...

https://arxiv.org/abs/2605.07485

03 CONF

LARAG: Link-Aware Retrieval Strategy for RAG Systems in Hyperlinked Technical Documentation

ArXiv CS.AI 50%

arXiv:2605.07517v1 Announce Type: cross Abstract: Retrieval-Augmented Generation (RAG) enhances the factual grounding of Large Language Models by conditioning their outputs on external documents. However, standard embedding-based retrievers treat naturally structured corpora,...

https://arxiv.org/abs/2605.07517

04 CONF

Implicit Preference Alignment for Human Image Animation

ArXiv CS.AI 50%

arXiv:2605.07545v1 Announce Type: cross Abstract: Human image animation has witnessed significant advancements, yet generating high-fidelity hand motions remains a persistent challenge due to their high degrees of freedom and motion complexity. While reinforcement learning from...

https://arxiv.org/abs/2605.07545

05 CONF

Revisiting Transformer Layer Parameterization Through Causal Energy Minimization

ArXiv CS.AI 50%

arXiv:2605.07588v1 Announce Type: cross Abstract: Transformer blocks typically combine multi-head attention (MHA) for token mixing with gated MLPs for token-wise feature transformation, yet many choices in their parameterization remain largely empirical. We introduce Causal...

https://arxiv.org/abs/2605.07588

06 CONF

Mathematical Reasoning via Intervention-Based Time-Series Causal Discovery Using LLMs as Concept Mastery Simulators

ArXiv CS.AI 50%

arXiv:2605.07600v1 Announce Type: cross Abstract: Recent methods for improving LLM mathematical reasoning, whether through MCTS-based test-time search or causal graph-guided knowledge injection, cannot identify which concepts causally contribute to a correct answer, as the...

https://arxiv.org/abs/2605.07600

07 CONF

N\"urnberg NLP at PsyDefDetect: Multi-Axis Voter Ensembles for Psychological Defence Mechanism Classification

ArXiv CS.AI 50%

arXiv:2605.07606v1 Announce Type: cross Abstract: Detecting levels of psychological defence mechanisms in supportive conversations is inherently ambiguous. In the PsyDefDetect shared task at BioNLP 2026 the eight positive defence categories share surface language and differ only...

https://arxiv.org/abs/2605.07606

08 CONF

LithoBench: Benchmarking Large Multimodal Models for Remote-Sensing Lithology Interpretation

ArXiv CS.AI 50%

arXiv:2605.07640v1 Announce Type: cross Abstract: Remote sensing lithology interpretation is fundamental to geological surveys, mineral exploration, and regional geological mapping. Unlike general land-cover recognition, lithology interpretation is a knowledge-intensive task...

https://arxiv.org/abs/2605.07640

09 CONF

DRIP-R: A Benchmark for Decision-Making and Reasoning Under Real-World Policy Ambiguity in the Retail Domain

ArXiv CS.AI 50%

arXiv:2605.07699v1 Announce Type: cross Abstract: LLM-based agents are increasingly deployed for routine but consequential tasks in real-world domains, where their behavior is governed by inherently ambiguous domain policies that admit multiple valid interpretations. Despite the...

https://arxiv.org/abs/2605.07699

10 **CONF**

Cross-Attention and Encoder-Decoder Transformers: A Logical Characterization

ArXiv CS.AI **50%**

arXiv:2605.07705v1 Announce Type: cross Abstract: We give a novel logical characterization of encoder-decoder transformers, the foundational architecture for LLMs that also sees use in various settings that benefit from cross-attention. We study such transformers over text in...

https://arxiv.org/abs/2605.07705

11 **CONF**

The AI-Native Large-Scale Agile Software Development Manifesto

ArXiv CS.AI **50%**

arXiv:2605.07717v1 Announce Type: cross Abstract: Despite the widespread adoption of agile methods, achieving true agility at scale remains elusive. Large-scale agile frameworks remain largely human-centric and manual, relying on coordination meetings, artifact synchronization,...

https://arxiv.org/abs/2605.07717

12 **CONF**

An Efficient Hybrid Sparse Attention with CPU-GPU Parallelism for Long-Context Inference

ArXiv CS.AI **50%**

arXiv:2605.07719v1 Announce Type: cross Abstract: Long-context inference increasingly operates over CPU-resident KV caches, either because decoding-time KV states exceed GPU memory capacity or because disaggregated prefill-decode systems place KV data in host memory. Although...

https://arxiv.org/abs/2605.07719

13 **CONF**

Memory-Efficient Looped Transformer: Decoupling Compute from Memory in Looped Language Models

ArXiv CS.AI **50%**

arXiv:2605.07721v1 Announce Type: cross Abstract: Recurrent LLM architectures have emerged as a promising approach for improving reasoning, as they enable multi-step computation in the embedding space without generating intermediate tokens. Models such as Ouro perform reasoning...

https://arxiv.org/abs/2605.07721

14 **CONF**

Curated Synthetic Data Doesn't Have to Collapse: A Theoretical Study of Generative Retraining with Pluralistic Preferences

ArXiv CS.AI **50%**

arXiv:2605.07724v1 Announce Type: cross Abstract: Recursive retraining of generative models poses a critical representation challenge: when synthetic outputs are curated based on a fixed reward signal, the model tends to collapse onto a narrow set of outputs that over-optimize...

https://arxiv.org/abs/2605.07724

15 **CONF**

Benchmarking EngGPT2-16B-A3B against Comparable Italian and International Open-source LLMs

ArXiv CS.AI **50%**

arXiv:2605.07731v1 Announce Type: cross Abstract: This report benchmarks the performance of ENGINEERING Ingegneria Informatica S.p.A.'s EngGPT2MoE-16B-A3B LLM, a 16B parameter Mixture of Experts (MoE) model with 3B active parameters. Performance is investigated across a wide...

https://arxiv.org/abs/2605.07731

16 CONF

POETS: Uncertainty-Aware LLM Optimization via Compute-Efficient Policy Ensembles

ArXiv CS.AI 50%

arXiv:2605.07775v1 Announce Type: cross Abstract: Balancing exploration and exploitation is a core challenge in sequential decision-making and black-box optimization. We introduce POETS ($\textbf{Po}$licy $\textbf{E}$nsembles for $\textbf{T}$hompson $\textbf{S}$ampling), a novel...

https://arxiv.org/abs/2605.07775

17 CONF

APEX: Assumption-free Projection-based Embedding eXamination Metric for Image Quality Assessment

ArXiv CS.AI 50%

arXiv:2605.07786v1 Announce Type: cross Abstract: As generative models achieve unprecedented visual quality, the gold standard for image evaluation remains traditional feature-distribution metrics (e.g., FID). However, these metrics are provably hindered by the closed-vocabulary...

https://arxiv.org/abs/2605.07786

18 CONF

Exact Is Easier: Credit Assignment for Cooperative LLM Agents

ArXiv CS.AI 50%

arXiv:2603.06859v2 Announce Type: replace-cross Abstract: Removing an agent from a cooperative team to measure its contribution seems natural, yet in multi-agent LLM systems this evaluation distorts the result it claims to measure. This failure is not isolated: learned critics,...

https://arxiv.org/abs/2603.06859

19 CONF

\mathsf{VISTA}: Decentralized Machine Learning in Adversary Dominated Environments

ArXiv CS.AI 50%

arXiv:2605.07841v1 Announce Type: cross Abstract: Decentralized machine learning often relies on outsourcing computations, such as gradient evaluations, to untrusted worker nodes. Existing robust aggregation methods can mitigate malicious behavior under honest-majority...

https://arxiv.org/abs/2605.07841

20 CONF

MatryoshkaLoRA: Learning Accurate Hierarchical Low-Rank Representations for LLM Fine-Tuning

ArXiv CS.AI 50%

arXiv:2605.07850v1 Announce Type: cross Abstract: With the rise in scale for deep learning models to billions of parameters, the computational cost of fine-tuning remains a significant barrier to deployment. While Low-Rank Adaptation (LoRA) has become the standard for...

https://arxiv.org/abs/2605.07850

21 CONF

On the Tradeoffs of On-Device Generative Models in Federated Predictive Maintenance Systems

ArXiv CS.AI 50%

arXiv:2605.07860v1 Announce Type: cross Abstract: Federated Learning (FL) has emerged as a promising paradigm for preserving client data ownership and control over distributed Internet of Things (IoT) environments. While discriminative models dominate most FL use cases, recent...

https://arxiv.org/abs/2605.07860

22 **CONF**

Video Understanding Reward Modeling: A Robust Benchmark and Performant Reward Models

ArXiv CS.AI **50%**

arXiv:2605.07872v1 Announce Type: cross Abstract: Multimodal reward models have advanced substantially in text and image domains, yet progress in video understanding reward modeling remains severely limited by the lack of robust evaluation benchmarks and high-quality preference...

https://arxiv.org/abs/2605.07872

23 **CONF**

CoCoReviewBench: A Completeness- and Correctness-Oriented Benchmark for AI Reviewers

ArXiv CS.AI **50%**

arXiv:2605.07905v1 Announce Type: cross Abstract: Despite the rapid development of AI reviewers, evaluating such systems remains challenging: metrics favor overlap with human reviews over correctness. However, since human reviews often cover only a subset of salient issues and...

https://arxiv.org/abs/2605.07905

24 **CONF**

Sycophantic AI makes human interaction feel more effortful and less satisfying over time

ArXiv CS.AI **50%**

arXiv:2605.07912v1 Announce Type: cross Abstract: Millions of people now turn to artificial intelligence (AI) systems for personal advice, guidance, and support. Such systems can be sycophantic, frequently affirming users' views and beliefs. Across five preregistered studies (N...

https://arxiv.org/abs/2605.07912

25 **CONF**

Exploring the non-convexity in machine learning using quantum-inspired optimization

ArXiv CS.AI **50%**

arXiv:2605.07947v1 Announce Type: cross Abstract: The escalating complexity of modern machine learning necessitates solving challenging non-convex optimization problems, particularly in high-dimensional regimes and scenarios contaminated by gross outliers. Traditional...

https://arxiv.org/abs/2605.07947

26 **CONF**

Where's the Plan? Locating Latent Planning in Language Models with Lightweight Mechanistic Interventions

ArXiv CS.AI **50%**

arXiv:2605.07984v1 Announce Type: cross Abstract: We study planning site formation in language models -- where internal representations of structurally-constrained future tokens form during the forward pass, and whether they causally drive generation. Using rhyming-couplet...

https://arxiv.org/abs/2605.07984

27 **CONF**

Dooly: Configuration-Agnostic, Redundancy-Aware Profiling for LLM Inference Simulation

ArXiv CS.AI **50%**

arXiv:2605.07985v1 Announce Type: cross Abstract: Selecting the optimal LLM inference configuration requires evaluation across hardware, serving engines, attention backends, and model architectures, since no single choice performs best across all workloads. Profile-based...

https://arxiv.org/abs/2605.07985

28 **CONF**

Towards Apples to Apples for AI Evaluations: From Real-World Use Cases to Evaluation Scenarios

ArXiv CS.AI 50%

arXiv:2605.07986v1 Announce Type: cross Abstract: AI measurement science has a wide variety of methodologies and measurements for comparing AI systems, resulting in what often appear to be "apples-to-oranges" comparisons across AI evaluations. To move toward "apples-to-apples"...

https://arxiv.org/abs/2605.07986

29 **CONF**

Tool Calling is Linearly Readable and Steerable in Language Models

ArXiv CS.AI 50%

arXiv:2605.07990v1 Announce Type: cross Abstract: When a tool-calling agent picks the wrong tool, the failure is invisible until execution: the email gets sent, the meeting gets missed. Probing 12 instruction-tuned models across Gemma 3, Qwen 3, Qwen 2.5, and Llama 3.1 (270M to...

https://arxiv.org/abs/2605.07990

30 **CONF**

Graph-Structured Hyperdimensional Computing for Data-Efficient and Explainable Process-Structure-Property Prediction

ArXiv CS.AI 50%

arXiv:2605.07999v1 Announce Type: cross Abstract: Multiphoton photoreduction enables high-fidelity fabrication of complex 3D microstructures, yet reliable process-structure-property (PSP) prediction remains difficult because the available data are sparse, heterogeneous, and...

https://arxiv.org/abs/2605.07999

31 **CONF**

Position: Mechanistic Interpretability Must Disclose Identification Assumptions for Causal Claims

ArXiv CS.AI 50%

arXiv:2605.08012v1 Announce Type: cross Abstract: Mechanistic interpretability papers increasingly use causal vocabulary: circuits, mediators, causal abstraction, monosemanticity. Such claims require explicit identification assumptions. A purposive audit of 10 papers across four...

https://arxiv.org/abs/2605.08012

32 **CONF**

Globally Optimal Training of Spiking Neural Networks via Parameter Reconstruction

ArXiv CS.AI 50%

arXiv:2605.08022v1 Announce Type: cross Abstract: Spiking Neural Networks (SNNs) have been proposed as biologically plausible and energy-efficient alternatives to conventional Artificial Neural Networks (ANNs). However, the training of SNN usually relies on surrogate gradients...

https://arxiv.org/abs/2605.08022

33 **CONF**

Beyond Pairs: Your Language Model is Secretly Optimizing a Preference Graph

ArXiv CS.AI 50%

arXiv:2605.08037v1 Announce Type: cross Abstract: Direct Preference Optimization (DPO) aligns language models using pairwise preference comparisons, offering a simple and effective alternative to Reinforcement Learning (RL) from human feedback. However, in many practical...

https://arxiv.org/abs/2605.08037

34 CONF

Fast Byte Latent Transformer

ArXiv CS.AI 50%

arXiv:2605.08044v1 Announce Type: cross Abstract: Recent byte-level language models (LMs) match the performance of token-level models without relying on subword vocabularies, yet their utility is limited by slow, byte-by-byte autoregressive generation. We address this bottleneck...

https://arxiv.org/abs/2605.08044

35 CONF

CA-SQL: Complexity-Aware Inference Time Reasoning for Text-to-SQL via Exploration and Compute Budget Allocation

ArXiv CS.AI 50%

arXiv:2605.08057v1 Announce Type: cross Abstract: While recent advancements in inference-time learning have improved LLM reasoning on Text-to-SQL tasks, current solutions still struggle to perform well on the most challenging tasks in the Bird-Bench (BIRD) benchmark. This is due...

https://arxiv.org/abs/2605.08057

36 CONF

The Memory Curse: How Expanded Recall Erodes Cooperative Intent in LLM Agents

ArXiv CS.AI 50%

arXiv:2605.08060v1 Announce Type: cross Abstract: Context window expansion is often treated as a straightforward capability upgrade for LLMs, but we find it systematically fails in multi-agent social dilemmas. Across 7 LLMs and 4 games over 500 rounds, expanding accessible...

https://arxiv.org/abs/2605.08060

37 CONF

EmambaIR: Efficient Visual State Space Model for Event-guided Image Reconstruction

ArXiv CS.AI 50%

arXiv:2605.08073v1 Announce Type: cross Abstract: Recent event-based image reconstruction methods predominantly rely on Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs) to process complementary event information. However, these architectures face fundamental...

https://arxiv.org/abs/2605.08073

38 CONF

LLM-Based Agents for Competitive Landscape Mapping in Drug Asset Due Diligence

ArXiv CS.AI 50%

arXiv:2508.16571v4 Announce Type: replace Abstract: In this paper, we describe and benchmark a competitor-discovery component used within an agentic AI system for fast drug asset due diligence. A competitor-discovery AI agent, given an indication, retrieves all drugs comprising...

https://arxiv.org/abs/2508.16571

39 CONF

Are LLM Agents Behaviorally Coherent? Latent Profiles for Social Simulation

ArXiv CS.AI 50%

arXiv:2509.03736v2 Announce Type: replace Abstract: The impressive capabilities of Large Language Models (LLMs) raise the possibility that synthetic agents can serve as substitutes for real participants in human-subject research. To evaluate this claim, prior research has...

https://arxiv.org/abs/2509.03736

40 CONF

Automated Evaluation can Distinguish the Good and Bad AI Responses to Patient Questions about Hospitalization

ArXiv CS.AI 50%

arXiv:2510.00436v2 Announce Type: replace Abstract: Automated approaches to answer patient-posed health questions are rising, but selecting among systems requires reliable evaluation. The current gold standard for evaluating the free-text artificial intelligence (AI)...

https://arxiv.org/abs/2510.00436

41 CONF

BEAVER: An Efficient Deterministic LLM Verifier

ArXiv CS.AI 50%

arXiv:2512.05439v2 Announce Type: replace Abstract: As large language models (LLMs) transition from research prototypes to production systems, practitioners often need reliable methods to verify model outputs and characterize tail risk for safe deployment. While sampling-based...

https://arxiv.org/abs/2512.05439

42 CONF

OASES: Outcome-Aligned Search-Evaluation Co-Training for Agentic Search

ArXiv CS.AI 50%

arXiv:2604.03675v2 Announce Type: replace Abstract: Agentic search enables language models to solve knowledge-intensive tasks by adaptively acquiring external evidence over multiple steps. Reinforcement learning with verifiable rewards (RLVR) has emerged as a widely adopted...

https://arxiv.org/abs/2604.03675

43 CONF

Emergent social transmission of model-based representations without inference

ArXiv CS.AI 50%

arXiv:2604.05777v2 Announce Type: replace Abstract: How do people acquire rich, flexible knowledge about their environment from others despite limited cognitive capacity? Humans are often thought to rely on computationally costly mentalizing, such as inferring others' beliefs....

https://arxiv.org/abs/2604.05777

44 CONF

PAIR-Former: Budgeted Relational Multi-Instance Learning for Functional miRNA Target Prediction

ArXiv CS.AI 50%

arXiv:2602.00465v3 Announce Type: replace-cross Abstract: Functional miRNA--mRNA targeting is a large-bag prediction problem where each transcript yields a heavy-tailed pool of candidate target sites (CTSs), yet only a pair-level label is observed. Prior methods use max-pooling...

https://arxiv.org/abs/2602.00465

45 CONF

On Time, Within Budget: Constraint-Driven Online Resource Allocation for Agentic Workflows

ArXiv CS.AI 50%

arXiv:2605.06110v2 Announce Type: replace Abstract: Agentic systems increasingly solve complex user requests by executing orchestrated workflows, where subtasks are assigned to specialized models or tools and coordinated according to their dependencies. While recent work...

https://arxiv.org/abs/2605.06110

46 **CONF**

Safactory: A Scalable Agentic Infrastructure for Training Trustworthy Autonomous Intelligence

ArXiv CS.AI 50%

arXiv:2605.06230v2 Announce Type: replace Abstract: As large models evolve from conversational assistants into autonomous agents, challenges increasingly arise from long-horizon decision making, tool use, and real environment interaction. Existing agenticinfrastructure remain...

https://arxiv.org/abs/2605.06230

47 **CONF**

Seeing Like an AI: How LLMs Apply (and Misapply) Wikipedia Neutrality Norms

ArXiv CS.AI 50%

arXiv:2407.04183v5 Announce Type: replace-cross Abstract: Large language models (LLMs) are trained on broad corpora and then used in communities with specialized norms. Is providing LLMs with community rules enough for models to follow these norms? We evaluate LLMs' capacity to...

https://arxiv.org/abs/2407.04183

48 **CONF**

Direction for Detection: A Survey of Automated Vulnerability Detection and all of its Pain Points

ArXiv CS.AI 50%

arXiv:2412.11194v2 Announce Type: replace-cross Abstract: Security vulnerabilities in software can have severe consequences; however, manual vulnerability detection is costly and does not scale, especially as agentic coding frameworks increase the rate of code production. Over...

https://arxiv.org/abs/2412.11194

49 **CONF**

Semantic Integrity Matters: Benchmarking and Preserving High-Density Reasoning in KV Cache Compression

ArXiv CS.AI 50%

arXiv:2502.01941v3 Announce Type: replace-cross Abstract: While Key-Value (KV) cache compression is essential for efficient LLM inference, current evaluations disproportionately focus on sparse retrieval tasks, potentially masking the degradation of High-Density Reasoning where...

https://arxiv.org/abs/2502.01941

50 **CONF**

Generalized Euler Logarithm and its Applications in Machine Learning: Natural Gradient, Backpropagation, Generalized EG, Mirror Descent and OLPS

ArXiv CS.AI 50%

arXiv:2502.17500v3 Announce Type: replace-cross Abstract: This paper investigates in depth the fundamental properties of the two-parameter generalized Euler logarithm and its inverse, the associated deformed (a,b)-exponential function. We systematically clarify the parameter...

https://arxiv.org/abs/2502.17500

51 **CONF**

Uncertainty Quantification for Prior-Data Fitted Networks using Martingale Posteriors

ArXiv CS.AI 50%

arXiv:2505.11325v4 Announce Type: replace-cross Abstract: Prior-data fitted networks (PFNs) have emerged as promising foundation models for prediction from tabular datasets, achieving state-of-the-art performance on small to moderate data sizes without tuning. While PFNs are...

https://arxiv.org/abs/2505.11325

52 CONF

HYPER: A Foundation Model for Inductive Link Prediction with Knowledge Hypergraphs

ArXiv CS.AI 50%

arXiv:2506.12362v3 Announce Type: replace-cross Abstract: Inductive link prediction with knowledge hypergraphs is the task of predicting missing hyperedges involving completely novel entities (i.e., nodes unseen during training). Existing methods for inductive link prediction...

https://arxiv.org/abs/2506.12362

53 CONF

SpikingBrain: Spiking Brain-inspired Large Models

ArXiv CS.AI 50%

arXiv:2509.05276v4 Announce Type: replace-cross Abstract: Mainstream Transformer-based large language models face major efficiency bottlenecks: training computation scales quadratically with sequence length, and inference memory grows linearly, limiting long-context processing....

https://arxiv.org/abs/2509.05276

54 CONF

A Computer Vision Pipeline for Individual-Level Behavior Analysis: Benchmarking on the Edinburgh Pig Dataset

ArXiv CS.AI 50%

arXiv:2509.12047v2 Announce Type: replace-cross Abstract: Animal behavior analysis plays a crucial role in understanding animal welfare, health status, and productivity in agricultural settings. However, traditional manual observation methods are time-consuming, subjective, and...

https://arxiv.org/abs/2509.12047

55 CONF

Frequency-Aware Model Parameter Explorer: A new attribution method for improving explainability

ArXiv CS.AI 50%

arXiv:2510.03245v2 Announce Type: replace-cross Abstract: State-of-the-art attribution methods rely on adversarial sample generation that applies an all-pass filter across the frequency spectrum, discarding fine-grained high-frequency information that is demonstrably important...

https://arxiv.org/abs/2510.03245

56 CONF

EvolveR: Self-Evolving LLM Agents through an Experience-Driven Lifecycle

ArXiv CS.AI 50%

arXiv:2510.16079v2 Announce Type: replace-cross Abstract: Current Large Language Model (LLM) agents show strong performance in tool use, but lack the crucial capability to systematically learn from their own experiences. While existing frameworks mainly focus on mitigating...

https://arxiv.org/abs/2510.16079

57 CONF

Is Chain-of-Thought Really Not Explainability? Chain-of-Thought Can Be Faithful without Hint Verbalization

ArXiv CS.AI 50%

arXiv:2512.23032v2 Announce Type: replace-cross Abstract: Recent work, using the Biasing Features metric, labels a CoT as unfaithful if it omits a prompt-injected hint that affected the prediction. We argue this metric adopts a narrow notion of faithfulness and confuses...

https://arxiv.org/abs/2512.23032

58 **CONF**

MirrorMark: Generalizable Mirrored Sampling for Multi-bit LLM Watermarking

ArXiv CS.AI 50%

arXiv:2601.22246v3 Announce Type: replace-cross Abstract: As large language models (LLMs) become integral to applications such as question answering and content creation, reliable content attribution has become increasingly important. Watermarking is a promising approach, but...

https://arxiv.org/abs/2601.22246

59 **CONF**

AdaCorrection: Adaptive Offset Cache Correction for Accurate Diffusion Transformers

ArXiv CS.AI 50%

arXiv:2602.13357v2 Announce Type: replace-cross Abstract: Diffusion Transformers (DiTs) achieve state-of-the-art performance in high-fidelity image and video generation but suffer from expensive inference due to their iterative denoising structure. While prior methods accelerate...

https://arxiv.org/abs/2602.13357

60 **CONF**

ScrapeGraphAI-100k: Dataset for Schema-Constrained LLM Generation

ArXiv CS.AI 50%

arXiv:2602.15189v2 Announce Type: replace-cross Abstract: Producing output that conforms to a specified JSON schema underlies tool use, structured extraction, and knowledge base construction in modern large language models. Despite this centrality, public datasets for the task...

https://arxiv.org/abs/2602.15189

61 **CONF**

Direction-Flipped Influence Audits Reveal Hidden Structure in Moral Choices of LLMs

ArXiv CS.AI 50%

arXiv:2602.22831v2 Announce Type: replace-cross Abstract: Moral benchmarks for LLMs typically score models on context-free prompts, implicitly treating the measured choice rate as stable. We test this assumption with a direction-flipped influence audit: for each scenario, we...

https://arxiv.org/abs/2602.22831

62 **CONF**

VDCook:DIY video data cook your MLLMs

ArXiv CS.AI 50%

arXiv:2603.05539v2 Announce Type: replace-cross Abstract: We introduce VDCook: a self-evolving video data operating system, a configurable video data construction platform for researchers and vertical domain teams. Users initiate data requests via natural language queries and...

https://arxiv.org/abs/2603.05539

63 **CONF**

Automatic Image-Level Morphological Trait Annotation for Organismal Images

ArXiv CS.AI 50%

arXiv:2604.01619v3 Announce Type: replace-cross Abstract: Morphological traits are physical characteristics of biological organisms that provide vital clues on how organisms interact with their environment. Yet extracting these traits remains a slow, expert-driven process,...

https://arxiv.org/abs/2604.01619

64 CONF

Governed Capability Evolution: Lifecycle-Time Compatibility Checking and Rollback for AI-Component-Based Systems, with Embodied Agents as Case Study

ArXiv CS.AI 50%

arXiv:2604.08059v4 Announce Type: replace-cross Abstract: Software systems built from versioned AI components increasingly need lifecycle-time governance: when a capability module evolves into a new version, the hosting system must decide whetmeher the new version may be...

https://arxiv.org/abs/2604.08059